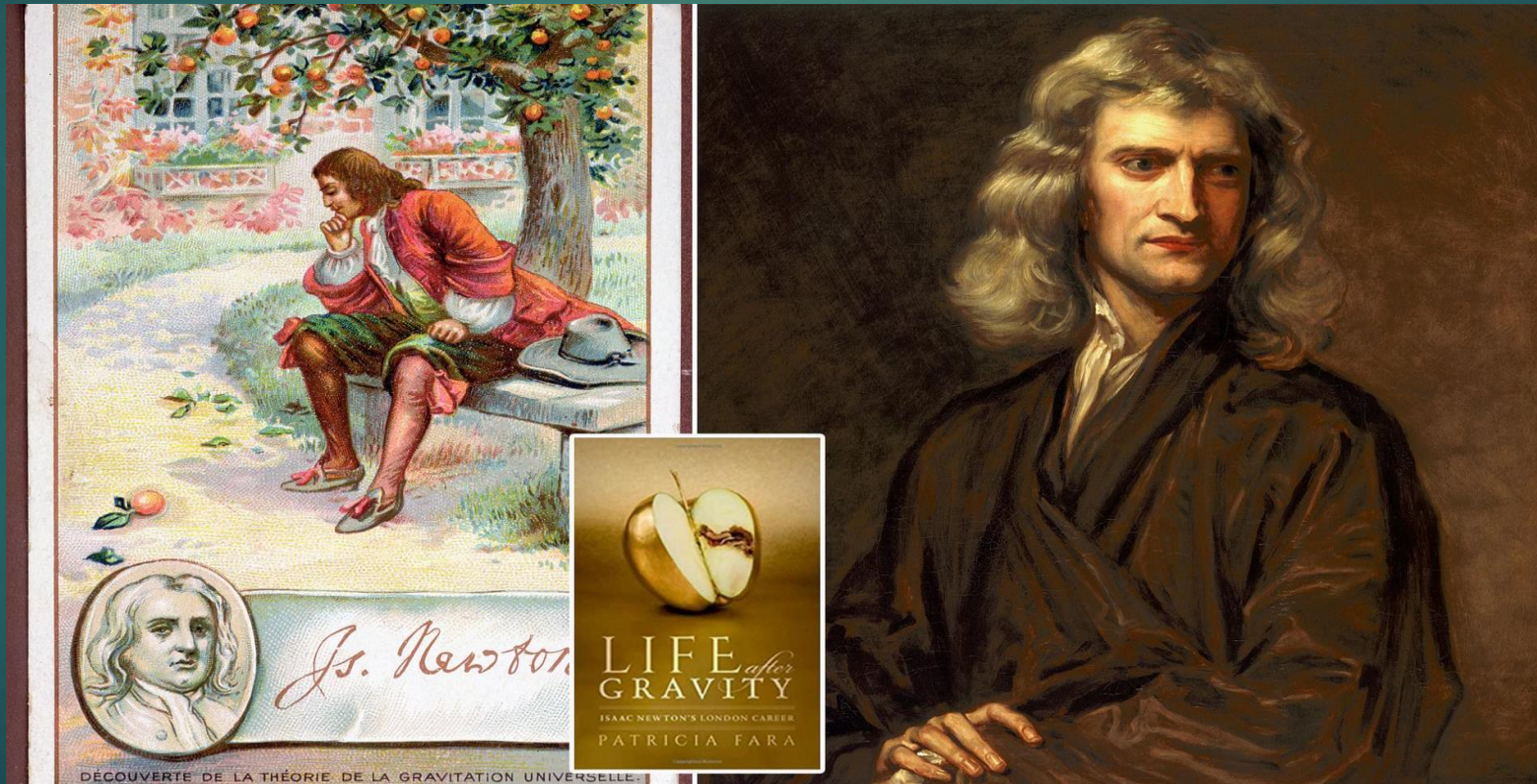


Experiment #(3)

Newton's Second Law

(Attwood Machine)

(قانون نيوتن الثاني)



Introduction

Newton's second law states that the **acceleration** of an object is directly proportional to the **net force** acting on it and inversely proportional to its **mass**. Newton's second law is written as $\sum \vec{F} = \vec{F}_{net} = m\vec{a}$. Here, we investigate the validity of Newton's second law in an experiment consisting of two parts. During the first part, we will use a variable force and a constant mass while in the second part, we will use a constant force and a variable mass.

Theory

In Newton's second law experiment (Atwood Machine) as shown in the figure

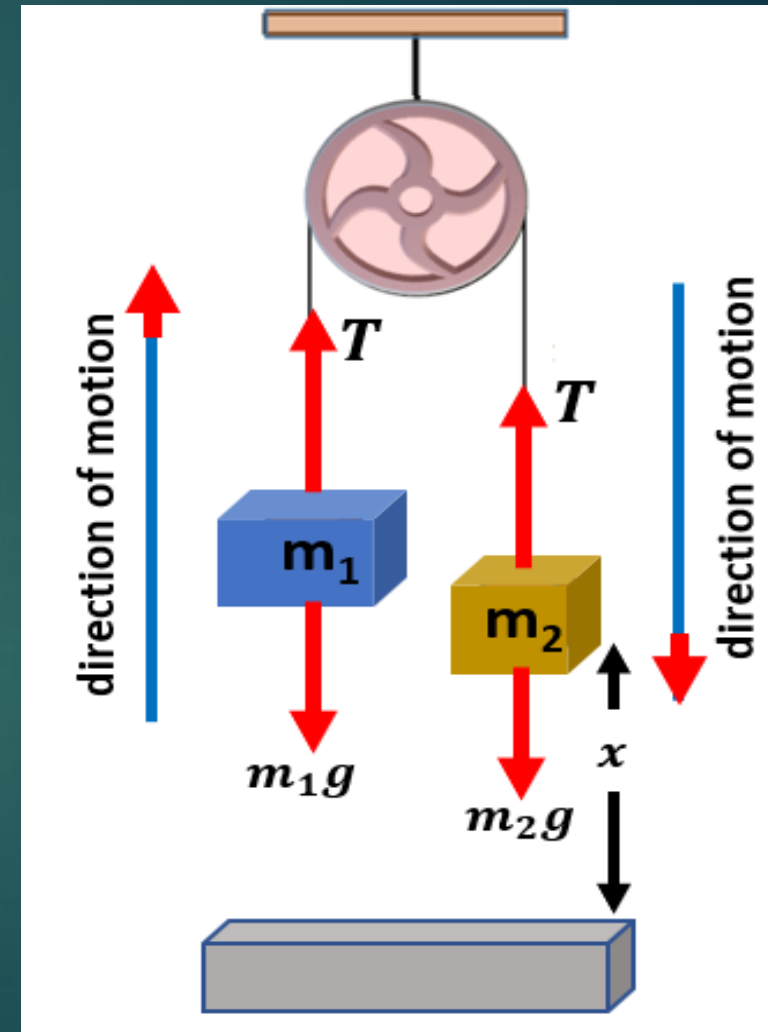
If $m_2 > m_1$ and the initial speed is zero

$$m_2 a = m_2 g - T$$

$$m_1 a = T - m_1 g$$



$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g$$



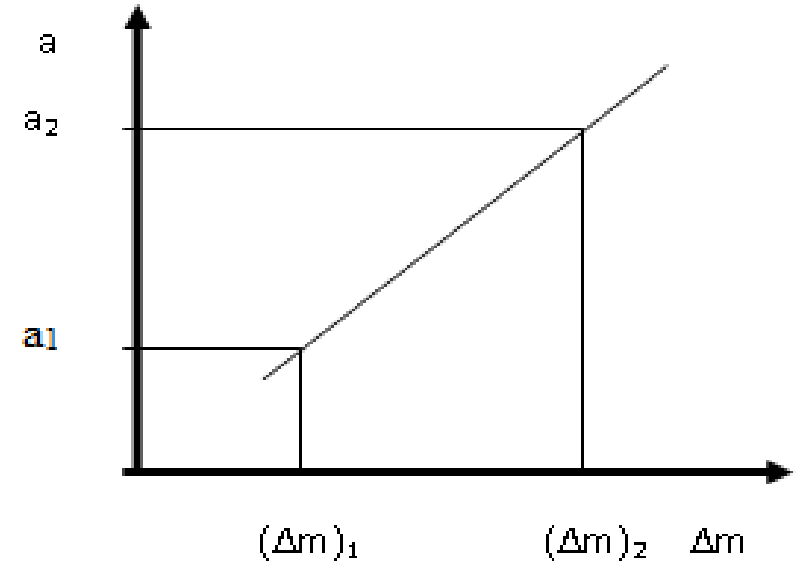
1- Constant Mass (ثبوت الكتلة)

$$m_1 + m_2 = \text{constant}$$

$$a = \left[\frac{g}{m_1 + m_2} \right] (m_2 - m_1)$$

$$\text{slope} = \frac{g}{m_1 + m_2}$$

$$g = (m_1 + m_2) \times \text{slope}$$



2: Constant Force (ثبوت القوة)

$$F_{\text{net}} = (m_1 - m_2)g = \text{constant},$$

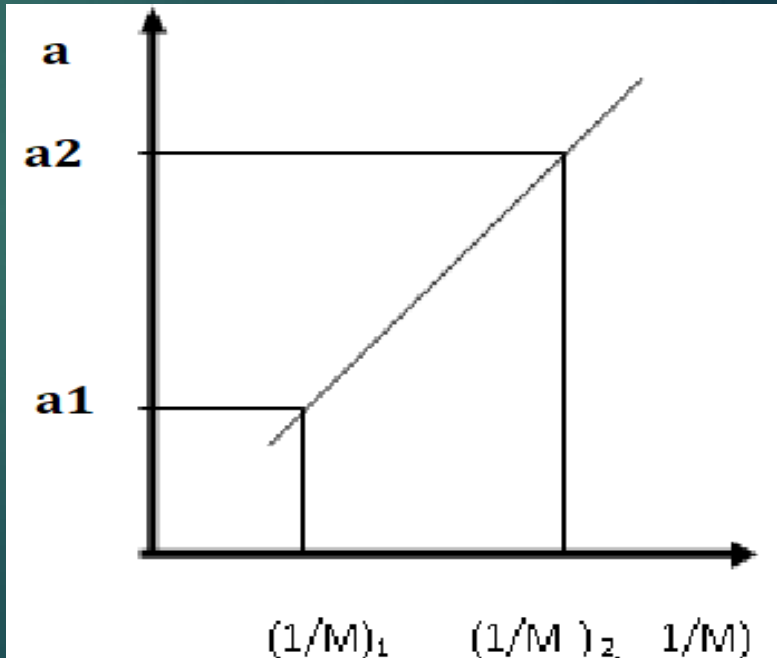
$$M = m_1 + m_2$$

$$m_2 - m_1 = \text{constant}$$

$$a = [(m_2 - m_1)g] \frac{1}{(m_1 + m_2)}$$

$$\text{slope} = (m_2 - m_1)g = F_{\text{net}}$$

$$g = \frac{\text{slope}}{(m_1 - m_2)}$$



Procedure Part 1: Constant Mass

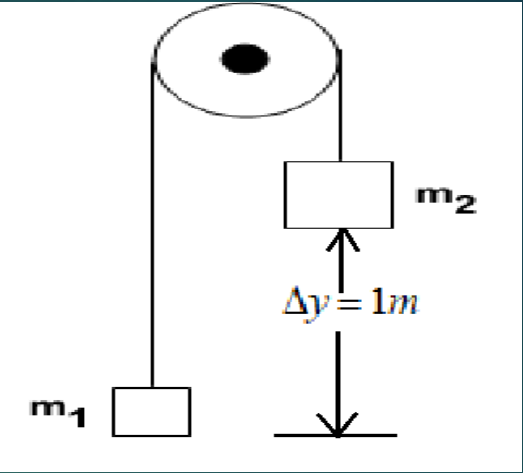
$m_1 + m_2 = \text{constant} = 200\text{g}$

- As a start, choose m_1 , and m_2 in such away that the total mass $m_2 + m_1 = 200\text{ g}$.
- Choose the hight of m_2 from the ground to be $x = 1\text{ m}$.
- Let m_2 falls down from rest and measure the time needed for m_2 to hit the ground.
- Repeat the trial three times and calculate the average value of the time needed by the m_2 to travel the distance x .
- Change the values of m_1 and m_2 in such away that the t otal mass $m_2 + m_1 = 200\text{ g} = \text{constant}$.
- Calculate the acceleration of the system using equation.

$\Delta y = v_0 t + \frac{1}{2} a t^2$ let $\Delta y = 1\text{m}$ and $v_0 = 0$

$a = \frac{2}{t^2}$

- Tabulate your results in the table shown below.



$x = 1\text{m}$				$M = m_1 + m_2 = 200\text{g}$					
Trial	m_1 (g)	m_2 (g)	$m_1 - m_2$ ()	t_1 ()	t_2 ()	t_3 ()	t_{avg} ()	t^2 ()	a ()
1	150	50							
2	140	60							
3	130	70							
4	120	80							
5	110	90							

Procedure Part 2: Constant Force

$F_{\text{net}} = (m_1 - m_2)g = \text{constant}$

$(m_1 - m_2) = 20g$

- As a start, choose m_1 , and m_2 in such away that the difference $m_2 - m_1 = 20\text{ g}$.
- Choose the hight of m_2 from the ground to be $x = 1\text{ m}$.
- Let m_2 falls down from rest and measure the time needed for m_2 to hit the ground.
- Repeat the trial three times and calculate the average value of the time needed by the m_2 to travel the distance x .
- Change the values of m_1 and m_2 in such away that the difference $m_2 - m_1 = \text{constant} = 20\text{ g}$.
- Calculate the acceleration of the system using equation .

$\Delta y = v_0t + \frac{1}{2}at^2$ let $\Delta y = 1m$ and $v_0 = 0$

$a = \frac{2}{t^2}$

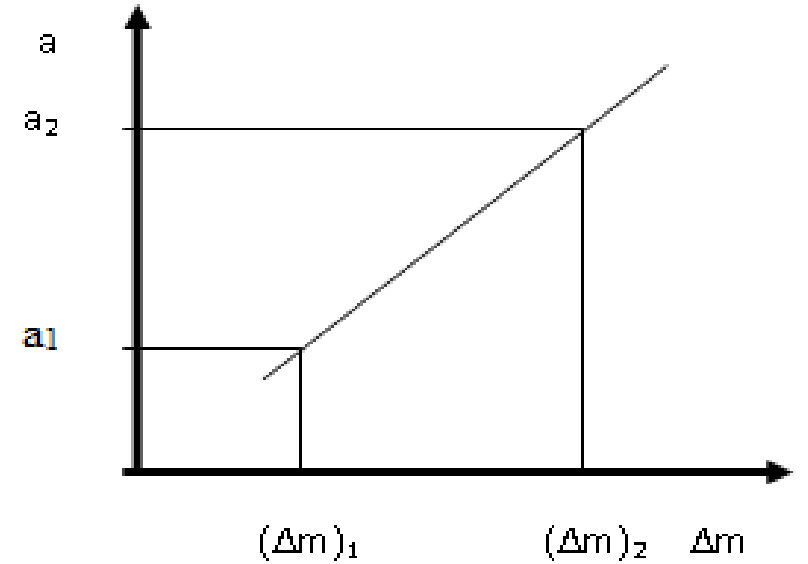
- Tabulate your results in the table shown below.

$x = 1\text{ m}$				$m_2 - m_1 = 20\text{ g}$					
Trial	m_1 (g)	m_2 (g)	$\frac{1}{m_1 + m_2}$	t_1 ()	t_2 ()	t_3 ()	t_{avg} ()	t^2 ()	a ()
1	70	50							
2	90	70							
3	110	90							
4	130	110							
5	150	130							

Analysis

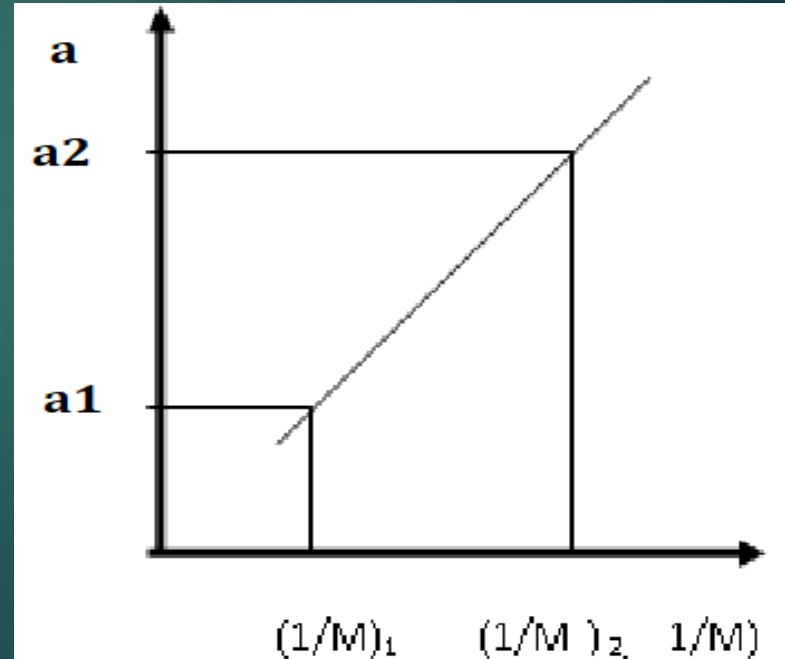
1- Constant Mass (ثبوت الكتلة)

- Plot the acceleration a versus $(m_2 - m_1)$ on a graph paper and draw the best straight line through the points.
- Find the slope. Use **equation 3** and the calculated value of the **slope** to calculate the earth's gravitational acceleration.
- Calculate the absolute and relative errors in the earth's gravitational acceleration.
- What are the possible sources of errors in this part?



2- Constant Force (ثبوت القوة)

- Plot the acceleration a versus $\frac{1}{m_1 + m_2}$ on a graph paper and draw the best straight line through the points.
- Find the slope. Use **equation 3** and the calculated value of the **slope** to calculate the earth's gravitational acceleration.
- Calculate the absolute and relative errors in the earth's gravitational acceleration.
- What are the possible sources of errors in this part?



THANK YOU
